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1 Introduction

1.1 Arithmetic in [0,infty]

1.2 Metric Space

Lemma 1.2.1

X is a metric space, $x \in X$, B_1, B_2 are twos balls in X. If $x \in B_1 \cap B_2$, then x is the center of an open ball $B \subseteq B_1 \cap B_2$.

Proof. We may assume that B_1, B_2 have different center. Let $y_1 \neq y_2$,

$$B_1 = \{x : d(x, y_1) < r_1\}, \quad B_2 = \{x : d(x, y_2) < r_2\}$$

Take $x \in B_1 \cap B_2$, we claim that there exists $r_0 > 0$ such that $B := \{y : d(x, y) < r_0\} \subseteq B_1 \cap B_2$. Otherwise , for all $r > 0$, there exists a point y_r such that the following two hold at the same time

1. $d(x, y_r) < r$
2. $d(y_r, y_1) \geq r_1$ or $d(y_r, y_2) \geq r_2$.

Thus

$$d(x, y_1) \geq d(y_r, y_1) - d(x, y_r) > r_1 - r, \quad \text{or} \quad d(x, y_2) > r_2 - r$$

The above shows that

$$r > \min\{r_1 - d(x, y_1), r_2 - d(x, y_2)\}, \quad \forall r > 0$$

which is a contradiction. □

1.3 Limits of Set Sequences

Definition 1.3.1 ▸ Limit Inferior and Limit Superior

Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of sets.

1. By the ***Limit Inferior of the sequence***, we mean

$$\liminf_{n\to\infty} A_n = \bigcup_{N=1}^{\infty}\bigcap_{n=N}^{\infty} A_n = \{x : \exists N_0 \in \mathbb{N}, \forall n > N_0, x \in A_n\}$$

2. By the ***Limit Superior of the sequence***,we mean

$$\limsup_{n\to\infty} A_n = \bigcap_{N=1}^{\infty}\bigcup_{n=N}^{\infty} A_n = \{x : \forall N \in \mathbb{N}, \exists n \geq N, x \in A_n\}$$

Note.

1. Limit Inferior 包含那些“最终稳定下来”的元素，即从某个点之后就永远属于序列中所有后续集合的元素。
2. Limit Superior 包含那些“反复出现”的元素，即在无限多个集合中出现的元素。

Proposition 1.3.2

For any sequence of sets $\{A_n\}$, it always holds that:

$$\liminf_{n\to\infty} A_n \subseteq \limsup_{n\to\infty} A_n$$

Proof sketch.
直觉上是显然的，因为“最终稳定下来”的元素一定会“反复出现”.

Proof. It is obvious by the $\{x : x \in P\}$ form representation of the Limit Inferior and Superior. □

Definition 1.3.3 ▸ Limit of a Set Sequence

If the limit inferior and limit superior of a set sequence $\{A_n\}_{n=1}^{\infty}$ are equal, i.e., $\liminf_{n\to\infty} A_n = \limsup_{n\to\infty} A_n$, then we say the ***limit*** of the sequence exists, and it is defined as:

$$\lim_{n\to\infty} A_n = \liminf_{n\to\infty} A_n = \limsup_{n\to\infty} A_n$$

Proposition 1.3.4

Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of sets.

1. If $\{A_n\}$ is an increasing sequence, then the limit of $\{A_n\}$ exists and

$$\lim_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} A_n$$

2. If $\{A_n\}$ is a decreasing sequence, then the limit of $\{A_n\}$ exists and

$$\lim_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

Proof. 1. If $\{A_n\}$ is increasing , then

$$\bigcap_{n=N}^{\infty} A_n = A_N$$

and

$$\bigcup_{n=N}^{\infty} A_n = \bigcup_{n=1}^{\infty} A_n, \forall N \in \mathbb{N}.$$

Thus

$$\liminf_{n\to\infty} A_n = \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} A_n = \bigcup_{N=1}^{\infty} A_N$$

and

$$\limsup_{n\to\infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} A_n$$

which are the same.

2. If $\{A_n\}$ is decreasing, then

$$\bigcap_{n=N}^{\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

and

$$\bigcup_{n=N}^{\infty} A_n = A_N$$

Thus

$$\liminf_{n\to\infty} A_n = \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} A_n = \bigcup_{N=1}^{\infty} \bigcap_{n=1}^{\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

and

$$\limsup_{n\to\infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n = \bigcap_{N=1}^{\infty} A_N$$

which are the same.

□

Proposition 1.3.5

For a sequence of sets $\{A_n\}_{n=1}^{\infty}$ and their corresponding sequence of indicator functions $\{\chi_{A_n}(x)\}_{n=1}^{\infty}$:

- $\chi_{\liminf_{n\to\infty} A_n}(x) = \liminf_{n\to\infty} \chi_{A_n}(x)$
- $\chi_{\limsup_{n\to\infty} A_n}(x) = \limsup_{n\to\infty} \chi_{A_n}(x)$
- $\lim_{n\to\infty} A_n$ exists, then $\chi_{\lim_{n\to\infty} A_n}(x) = \lim_{n\to\infty} \chi_{A_n}(x)$

Proof sketch.

- 若 x 在 limit inferior 里面，则 x 是“最终稳定”的，$\chi_{A_n}(x)$ 是关于 n “最终”恒为 1 的.
- 若 x 在 limit superior 里面，则 x 是“反复出现”的，即相当于 N 多大，总会出现之后的某个 n 使得 $\chi_{A_n}(x) = 1$.
- 当极限存在时，函数列 $\left\{\chi_{A_n}(x)\right\}_{n=1}^{\infty}$ 的 limit inferior 和 supperior 根据上两条相等，等于极限集合的 χ.

2

Abstract Integration

2.1 The Concept of Measurbility

Definition 2.1.1

1. A collection $\mathfrak{M}$ of subsets of a set X is said to be a ***σ-algebra*** in X if $\mathfrak{M}$ has the following properties:
   (a) $X \in \mathfrak{M}$.
   (b) If $A \in \mathfrak{M}$, then $A^c \in \mathfrak{M}$.
   (c) If $A = \bigcup_{n=1}^{\infty} A_n$ and $A_n \in \mathfrak{M}$ for $n = 1, 2, 3, \cdots$, then $A \in \mathfrak{M}$.
2. If $\mathfrak{M}$ is a σ-algebra in X, then X is called a ***measurable*** provided that $f^{-1}(V)$ is measurable set in X for every open set V in Y.
3. If X is a measurable space, Y is a topological space, and f is a mapping of X into Y, then f is said to be ***measurable*** provided that $f^{-1}(V)$ is a measurable set in X for every open set V in Y.

Note.
对比 topological space 的定义, σ-algebra 的定义是对称的, 即可测集的补集仍是可测的, 而对于拓扑空间, 开集的补集不是开集, 而是被定义为了闭集这样的对象.

Lemma 2.1.2

1. Let X be a measurable space , Y, Z be topological spaces. If $f : X \to Y$ is measurable , and if $g : Y \to Z$ is continuous, then $g \circ f : X \to Z$ is measurable.
2. Let u, v be real measurable functions on a measurable space X, let Φ be a continuous mapping of the plane into a topological space Y, and define
$$h(x) = \Phi(u(x), v(x))$$
for $x \in X$. Then $h : X \to Y$ is measurable.

Proof. 1. Take any open subset $V \subseteq Z$. Then $g^{-1}(V)$ is an open subset of

Y . Since f is measurable,

$$(g \circ f)^{-1}(V) = f^{-1}\left(g^{-1}(V)\right)$$

is a mearuable set.

2. Let $V \subseteq Y$ be a open subset. Then $\Phi^{-1}(V)$ is a open subset of $[-\infty, \infty]^2$. There exists countably many open sets of $[-\infty, \infty]$: $U_i, V_i, i \in \mathbb{N}$, such that

$$\Phi^{-1}(V) = \bigcup_i U_i \times V_i$$

Thus

$$\begin{aligned} h^{-1}(V) &= (u, v)^{-1}\left(\bigcup_i U_i \times V_i\right) \\ &= \bigcup_i (u, v)^{-1}(U_i \times V_i) \\ &= \bigcup_i \left(u^{-1}(U_i) \cap v^{-1}(V_i)\right) \end{aligned}$$

which is a measurable set.

□

Corollary 2.1.3

Let X be a measurable space. The following propositions hold

1. If $f = u + iv$, where u and v are real measurable functions on X, then f is a complex measurable function on X.
2. If $f = u + iv$ is a complex measurable function on X, then u, v, and $|f|$ are real measurable functions on X.
3. If f and g are complex measurable functions on X, then so are $f + g$ and fg.
4. If E is a measurable set in X and if

$$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$$

then χ_E is a measurable function.

5. If f is a complex measurable function on X , there is a complex measurable function α on X such that $|\alpha| = 1$ and $f = \alpha |f|$.

Theorem 2.1.4

If $\mathscr{F}$ is any collection of subsets of X , there exists a smallest σ-algebra $\mathfrak{M}^*$ such that $\mathscr{F} \subseteq \mathfrak{M}^*$.

Definition 2.1.5 ▶ Borel

Let X be a topological space.

1. By ***Borel σ-algebra*** , we mean the smallest σ-algebra $\mathscr{B}$ in X such that every open set in X belongs to $\mathscr{B}$. The members of $\mathscr{B}$ are called the ***Borel sets*** of X.
2. All countable unions of closed sets and all countable intersections of open sets are Borel sets, which we called ***F_σ's and G_δ's***, respectively.
3. By a ***Borel function***, we mean a measurable function on the measurable space $(X, \mathscr{B})$.

Remark.

1. The letters F and G were used for colsed and opensets,respectively, and σ refers to union, δ to intersection.
2. A continuous function is Borel-measurable, since the preimage of any open set is open and therefore a Borel set.

Theorem 2.1.6

Suppose $\mathfrak{M}$ is a σ-algbera in X ,and Y is a topological space. Let f map X into Y.

1. If Ω is the collection of all sets $E \subseteq Y$ such that $f^{-1}(E) \in \mathfrak{M}$, then Ω is a σ- algebra in Y.
2. If f is measurable and E is a Borel set in Y , then $f^{-1}(E) \subseteq \mathfrak{M}$.
3. If $Y = [-\infty, \infty]$ and $f^{-1}((\alpha, \infty]) \subseteq \mathfrak{M}$ for every real α, then f is measurable .
4. If f is measurable, if Z is topological space, if $g : Y \to Z$ is a Borel mapping, and if $h = g \circ f$, then $h : X \to Z$ is measurable.

Theorem 2.1.7

If $f_n : X \to [-\infty, \infty]$ is measurable,for $\neq 1, 2, 3, \cdots$, and

$$g = \sup_{n \geq 1} f_n, \quad h = \lim_{n \to \infty} \sup f_n,$$

then g and h are measurable.

Corollary 2.1.8

1. The limit of every pointwise convergent sequence of complex measurable functions is measurable.
2. If f and g are measurable(with range in $[-\infty, \infty]$), then so are $\max\{f, g\}$ and $\min\{f, g\}$. In particular , this is true of the functions

$$f^+ = \max\{f, 0\}, \quad f^- = -\min\{f, 0\}.$$

which are called the ***positive part*** and ***negative part*** of f , respectively.

Remark.

1. There are the standrd representation

$$|f| = f^+ + f^-, \quad f = f^+ - f^-$$

2. And easy but (may) useful observation is : If $f = g - h, g \geq 0, h \geq 0$, then $f^+ \leq g$ and $f^- \leq h$.

2.2 Simple Functions

Definition 2.2.1

A complex function s on a measurable space X whose range consists of only finitely many points will be called a ***simple function***. Among these are the nonnegative simple functions, whose range is a finite subset of $[0, \infty)$.
Specifically, if $\alpha_1, \cdots, \alpha_n$ are distinct values of a simple function s, and if we set $A_i = \{x : s(x) = \alpha_i\}$, then clearly

$$s = \sum_{i=1}^{n} \alpha_i \chi_{A_i}.$$

Where χ_{A_i} is the characteristic function of A_i.

Remark.
- Here, we explicity exclude ∞ from the values of a simple function.
- It is clear that s is mearuable if and only if each of the sets A_i is measurable.

Theorem 2.2.2

Let $f : X \to [0, \infty]$ be measurable space. There exists simple measurable functions s_n on X such that
1. $0 \le s_1 \le s_2 \le \cdots \le f$.
2. $s_n(x) \to f(x)$ as $n \to \infty$, for every $x \in X$.

Proof sketch.
We construst a sequence of Borel simple functions $\{\varphi_n(x)\}$ to act as an ***identify***. 当 n 增大的同时, 我们同时让 $\varphi_n(x)$ 的单位逼近精度和单位逼近范围随着 n 提升. 并且在舍弃误差时, 总是向下取整, 使得该单位逼近是自下而上的.

Proof. For every $x \in [0, \infty]$, and for every $n \in \mathbb{N}$, there exists a unique integer $k_n(x)$, such that

$$k_n(x) 2^{-n} \le x < (k_n(x) + 1) 2^{-n}$$

For every $n \in \mathbb{N}$, we define

$$\varphi_n(x) = \begin{cases} k_n(x) 2^{-n}, & 0 \le x \le n \\ n, & x \ge n \end{cases}$$

Each $\varphi_n(x)$ is then a Borel simple function. It is not hard to show that $0 \le \varphi_1 \le \varphi_2 \cdots \le \cdots \le \mathrm{Id}$. For each n, we define

$$s_n(x) := (\varphi_n \circ f)(x)$$

Then $\{s_n\}$ is a suquence of simple measurable functions such that $0 \le s_1 \le s_2 \le \cdots \le f$. Since $\lim_{n\to\infty} \varphi_n(x) = x$, then $\lim_{n\to\infty} s_n(x) = f(x)$. $\square$

2.3 Measure

Definition 2.3.1 ▶ Measure and Measure Space

(a) A ***positive measure*** is a function μ, defined on a σ-algebra $\mathfrak{M}$, whose range is in $[0, \infty]$ and which is ***countably additive***. This means that if $\{A_i\}$ is a disjoint countable collection of members of $\mathfrak{M}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu\left(A_i\right).$$

To avoid trivialities, we shall also assume that $\mu(A) < \infty$ for at least one $A \in \mathfrak{M}$.

(b) A ***measure space*** is a measurable space which has a positive measure defined on the σ-algebra of its measurable sets.

(c) A ***complex measure*** is a complex-valued countably additive function defined on a σ-algebra.

Theorem 2.3.2

Let μ be a positive measure on a σ-algebra $\mathfrak{M}$. Then

1. $\mu(\varnothing) = 0$.
2. $\mu(A_1 \cup \cdots \cup A_n) = \mu(A_1) + \cdots + \mu(A_n)$ if $A_1, \ldots, A_n$ are pairwise disjoint members of $\mathfrak{M}$.
3. $A \subset B$ implies $\mu(A) \leq \mu(B)$ if $A \in \mathfrak{M}, B \in \mathfrak{M}$.
4. $\mu(A_n) \to \mu(A)$ as $n \to \infty$ if $A = \bigcup_{n=1}^{\infty} A_n, A_n \in \mathfrak{M}$, and
$$A_1 \subset A_2 \subset A_3 \subset \ldots$$
5. $\mu(A_n) \to \mu(A)$ as $n \to \infty$ if $A = \bigcap_{n=1}^{\infty} A_n, A_n \in \mathfrak{M}$,
$$A_1 \supset A_2 \supset A_3 \supset \ldots,$$
and $\mu(A_1)$ is finite.

Proof. 1. Take $A \in \mathfrak{M}$ such that $\mu\left(A\right) < \infty$. [1] And let $A_2 = A_3 = \cdots = \varnothing$, then $\mu\left(\varnothing\right) > 0$ leads to a contradition to the countably additive.

2. Take $A_{n+1} = A_{n+2} = \cdots = \varnothing$.

[1] That is what we supposed at the definition of measure.

3. Note that $B = (B \setminus A) \cup A$, then by additivity
$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$$

4. Let $A_0 = \varnothing$, and let $B_n = A_n \setminus A_{n-1}$ for all $n \in \mathbb{N}$. Then $B_1, \cdots, B_n$ are pairwise disjoint members of $\mathfrak{M}$ such that $A = \bigcup_{n=1}^{\infty} B_n$. We have
$$\mu(A) = \sum_{n=1}^{\infty} \mu(B_n) = \sum_{n=1}^{\infty} \mu(A_n \setminus A_{n-1})$$
If one of the $\mu(A_n)$ is ∞, then $\lim_{n\to\infty} \mu(A_n)$ and $\mu(A)$ both are ∞ . Otherwise, we have
$$\mu(A_n \setminus A_{n-1}) = \mu(A_n) - \mu(A_{n-1}), \quad \forall n \mathbb{N}$$
Thus
$$\mu(A) = \sum_{n=1}^{\infty} \mu(A_n \setminus A_{n-1}) = \lim_{n\to\infty} \mu(A_n)$$

5. Let $B_n = A_n \setminus A_{n+1}$ for all $n \in \mathbb{N}$. $B_1, \cdots, B_n$ are pariwise disjoint members of $\mathfrak{M}$ with finite measure, such that
$$A_1 \setminus A = A_1 \setminus \left(\bigcap_{n=1}^{\infty} A_n \right) = \bigcup_{n=1}^{\infty} (A_1 \setminus A_n) = \bigcup_{n=1}^{\infty} \left(\bigcup_{k=1}^{n} B_k \right) = \bigcup_{n=1}^{\infty} B_n$$
. We have
$$\mu(A_1 \setminus A) = \mu\left(\bigcup_{n=1}^{\infty} B_n \right),$$
where the RHS is $\mu(A_1) - \mu(A)$, and the LHS is
$$\sum_{n=1}^{\infty} (\mu(A_n) - \mu(A_{n+1})) = \mu(A_1) - \lim_{n\to\infty} \mu(A_{n+1})$$
Since $\mu(A_1) < \infty$, we have
$$\mu(A) = \lim_{n\to\infty} \mu(A_n)$$

$\square$

Example 2.3.3

1. For any $E \subset X$, where X is any set, define $\mu(E) = \infty$ if E is an infinite set, and let $\mu(E)$ be the number of points in E if E is finite. This μ is called the ***counting measure*** on X.
2. Fix $x_0 \in X$, define $\mu(E) = 1$ if $x_0 \in E$ and $\mu(E) = 0$ if $x_0 \notin E$, for any $E \subset X$. This μ may be called the ***unit mass concentrated at*** x_0.
3. Let μ be the counting measure on the set $\{1, 2, 3, \dots\}$, let $A_n = \{n, n+1, n+2, \dots\}$. Then $\bigcap A_n = \varnothing$ but $\mu(A_n) = \infty$ for $n = 1, 2, 3, \dots$. This shows that the hypothesis

$$\mu(A_1) < \infty$$

is not superfluous in Theorem 2.3.2(5).

2.4 Integration of Positive Funtions

Definition 2.4.1 ▶ Integration of Positive Funtions

1. If $s : X \to [0, \infty)$ is a measurable simple function, of the form

$$s = \sum_{i=1}^{n} \alpha_i \chi_{A_i},$$

where $\alpha_1, \dots, \alpha_n$ are the distinct values of s, and if $E \in \mathfrak{M}$, we define

$$\int_E s \, d\mu = \sum_{i=1}^{n} \alpha_i \mu(A_i \cap E).$$

The convention $0 \cdot \infty = 0$ is used here; it may happen that $\alpha_i = 0$ for some i and that $\mu(A_i \cap E) = \infty$.
2. If $f : X \to [0, \infty]$ is measurable, and $E \in \mathfrak{M}$, we define

$$\int_E f \, d\mu = \sup \int_E s \, d\mu,$$

the supremum being taken over all simple measurable functions s such that $0 \leq s \leq f$. The left member above is called the ***Lebesgue integral*** of f over E, with respect to the measure μ. It is a number in $[0, \infty]$.

Remark.
We apparently have two definitions for $\int_E f \, d\mu$ if f is simple, they are the same.

Corollary 2.4.2

(a) If $0 \le f \le g$, then $\int_E f \, d\mu \le \int_E g \, d\mu$.
(b) If $A \subset B$ and $f \ge 0$, then $\int_A f \, d\mu \le \int_B f \, d\mu$.
(c) If $f \ge 0$ and c is a constant, $0 \le c < \infty$, then

$$\int_E cf \, d\mu = c \int_E f \, d\mu.$$

(d) If $f(x) = 0$ for all $x \in E$, then $\int_E f \, d\mu = 0$, even if $\mu(E) = \infty$.
(e) If $\mu(E) = 0$, then $\int_E f \, d\mu = 0$, even if $f(x) = \infty$ for every $x \in E$.
(f) If $f \ge 0$, then $\int_E f \, d\mu = \int_X \chi_E f \, d\mu$.

Remark.
We can also regard the $\int_E f \, d\mu$ as the restricted function $f|_E$ integrates on the induced measure space E.

Proposition 2.4.3

Let s and t be nonnegative measurable simple functions on X. For $E \in \mathfrak{M}$, define

$$\varphi(E) = \int_E s \, d\mu.$$

Then φ is a measure on $\mathfrak{M}$. Also

$$\int_X (s+t) \, d\mu = \int_X s \, d\mu + \int_X t \, d\mu.$$

Note.
$\varphi(E) = \int_E s \, d\mu$ 就是依据 s 对测度 μ 进行一个有限的算数的加权. 自然地也就成为一个测度.

Proof. If $E_1, E_2, \cdots$ are disjoint members of $\mathfrak{M}$ whose union is E, the countable additivity of μ shows that

$$\begin{aligned}\varphi(E) &= \sum_{i=1}^{n} \alpha_i \mu(A_i \cap E) \\ &= \sum_{i=1}^{n} \sum_{j=0}^{\infty} \alpha_i \mu\left(A_i \cap E_j\right) \\ &= \sum_{j=0}^{\infty} \sum_{i=1}^{n} \alpha_i \mu\left(A_i \cap E_j\right) \\ &= \sum_{j=0}^{\infty} \varphi\left(E_j\right)\end{aligned}$$

Also, $\varphi(\varnothing) \neq \infty$, μ is not identically ∞.

Next, let $\beta_1, \cdots, \beta_m$ be distinct values of t. And let $B_j := \left\{x : t(x) = \beta_j\right\}$. Then

$$\begin{aligned}\int_X (s+t)\, \mathrm{d}\mu &= \sum_{i,j} \int_{A_i \cap B_j} (s+t)\, \mathrm{d}\mu \\ &= \sum_{i,j} \alpha_i \mu\left(A_i \cap B_j\right) + \sum_{i,j} \beta_j \mu\left(A_i \cap B_j\right) \\ &= \sum_i \alpha_i \mu(A_i) + \sum_j \beta_j \mu\left(B_j\right) \\ &= \int_X s\, \mathrm{d}\mu + \int_X t\, \mathrm{d}\mu\end{aligned}$$

□

Theorem 2.4.4 ▸ Lebesgue's Monotone Convergence Theorem

Let $\{f_n\}$ be a sequence of measurable functions on X, and suppose that
(a) $0 \leq f_1(x) \leq f_2(x) \leq \cdots \leq \infty$ for every $x \in X$,
(b) $f_n(x) \to f(x)$ as $n \to \infty$, for every $x \in X$.
Then f is measurable, and

$$\int_X f_n \, d\mu \to \int_X f \, d\mu \quad \text{as } n \to \infty.$$

Proof. Since $\int_X f_n \,\mathrm{d}\mu \le \int_X f_{n+1} \,\mathrm{d}\mu$ for all n, there exists an $\alpha > 0$ such that

$$\int_X f_n \,\mathrm{d}\mu \to \alpha, \quad \text{as } n \to \infty$$

By Theorem 2.1.7, f is measurable, then by $\int_X f_n \,\mathrm{d}\mu \le \int_X f \,\mathrm{d}\mu \;\forall n$,

$$\alpha \le \int_X f \,\mathrm{d}\mu$$

Take any simple function $0 \le s \le f$, and $0 \le c < 1$. Define

$$E_n^s := \{x \in X : f_n(x) \ge cs(x)\}$$

Then since $\varphi(E) := \int_E s \,\mathrm{d}\mu$ is a measure ,

$$\alpha \ge \lim_{n\to\infty} \int_X f_n \,\mathrm{d}\mu \ge \lim_{n\to\infty} \int_{E_n^s} f_n \,\mathrm{d}\mu \ge c \lim_{n\to\infty} \int_{E_n^s} s \,\mathrm{d}\mu = c \lim_{n\to\infty} \varphi(E_n^s) = c\varphi(X)$$

That is

$$\alpha \ge c \int_X s \,\mathrm{d}\mu$$

Since s is arbitrary,

$$\alpha \ge c \int_X f \,\mathrm{d}\mu$$

and since c is arbitrary ,

$$\alpha \ge \int_X f \,\mathrm{d}\mu$$

which completes the proof. □

Corollary 2.4.5

Let $f, g : X \to [0, \infty]$ be mearuable functions on X . Then $f + g$ is mearuable and

$$\int_X (f+g) \,\mathrm{d}\mu = \int_X f \,\mathrm{d}\mu + \int_X g \,\mathrm{d}\mu$$

Proof. Let $\{s_n\}, \{s'_n\}$ be simple function sequences in Theorem 2.2.2 of f, g , respectively . Then by the theorem above and the Proposition 2.4.3

$$\int_X (f+g)\,d\mu = \lim_{n\to\infty}\int_X (s_n + s_n')\,d\mu = \lim_{n\to\infty}\int_X s_n\,d\mu + \lim_{n\to\infty}\int_X s_n'\,d\mu$$
$$= \int_X f\,d\mu + \int_g d\mu$$

□

Corollary 2.4.6

If $f_n : X \to [0,\infty]$ is ***measurable***, for $n = 1, 2, 3, \ldots$, and

$$f(x) = \sum_{n=1}^{\infty} f_n(x) \quad (x \in X),$$

then

$$\int_X f\,d\mu = \sum_{n=1}^{\infty}\int_X f_n\,d\mu.$$

Proof. Let $g_n(x) = \sum_{k=1}^{n} f_k(x)$. Then $g_1 \le g_2 \le \cdots \le \infty$ and $f = \lim_{n\to\infty} g_n(x)$. Applying Monotone Convergence Theorem to $\{g_n\}$ and f , we have

$$\int_X f\,d\mu = \lim_{n\to\infty}\int_X \sum_{k=1}^{n} f_k(x)\,d\mu = \lim_{n\to\infty}\sum_{k=1}^{n}\int_X f_k(x)\,d\mu = \sum_{n=1}^{\infty}\int_X f_n\,d\mu$$

□

Corollary 2.4.7

If $a_{ij} \ge 0$ for i and $j = 1, 2, 3, \ldots$, then

$$\sum_{i=1}^{\infty}\sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty}\sum_{i=1}^{\infty} a_{ij}.$$

Proof. Let μ be counting measure on $\mathbb{Z}$. Let

$$f_i(j) = a_{ij}, \quad f(k) = \sum_{i=1}^{\infty} f_i(k) = \sum_{i=1}^{\infty} a_{ik}$$

By Monotone Convergence Theorem , for any measurable g ,

$$\int_{\mathbb{Z}} g \, \mathrm{d}\mu = \int_{\mathbb{Z}} \lim_{n\to\infty} \chi(\{1, \cdots, n\}) g \, \mathrm{d}\mu = \lim_{n\to\infty} \int_{\mathbb{Z}} \chi(\{1, \cdots, n\}) g \, \mathrm{d}\mu = \sum_{k=\infty}^{n} g(k)$$

Thus

$$\int_{\mathbb{Z}} f \, \mathrm{d}\mu = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}$$

And we have

$$\sum_{i=1}^{\infty} \int_X f_i \, \mathrm{d}\mu = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij}$$

The above corollary shows that

$$\int_{\mathbb{Z}} f \, \mathrm{d}\mu = \sum_{i=1}^{\infty} \int_X f_i \, \mathrm{d}\mu$$

That is

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}$$

□

Lemma 2.4.8 ▶ Fatou's Lemma

If $f_n : X \to [0, \infty]$ is ***measurable***, for each positive integer n, then

$$\int_X \left(\lim_{n\to\infty} \inf f_n \right) d\mu \leq \lim_{n\to\infty} \inf \int_X f_n d\mu. \tag{1}$$

Note.
有了 Monotone Convergence Theorem 之后, 关键的不等式就剩下

$$\int_X \inf_{n\geq k} f_n \,\mathrm{d}\mu \leq \inf_{n\geq k} \int_X f_n \,\mathrm{d}\mu$$

它的含义是局部最小的积分小于等于积分的最小.

Proof. Let

$$g_k(x) = \inf_{n\geq k} f_n(x)$$

Then

$$\lim_{n\to\infty} \inf f_n = \lim_{n\to\infty} g_n =: g$$

where $\{g_k\}, g$ are all measurable . It is obvious that $\{g_k\}$ is increasing . Thus

$$\lim_{n\to\infty} \int_X g_n \,\mathrm{d}\mu = \int_X g \,\mathrm{d}\mu = \int_X \left(\lim_{n\to\infty} \inf f_n\right) \mathrm{d}\mu$$

Only need to show that

$$\int_X \inf_{n\geq k} f_n \,\mathrm{d}\mu \leq \inf_{n\geq k} \int_X f_n \,\mathrm{d}\mu$$

which is true since

$$\int_X \inf_{n\geq k} f_n \,\mathrm{d}\mu \leq \int_X f_n \,\mathrm{d}\mu, \text{for all } n \geq k$$

□

Theorem 2.4.9

Suppose $f : X \to [0, \infty]$ is measurable, and

$$\varphi(E) = \int_E f \, d\mu \quad (E \in \mathfrak{M}).$$

Then φ is a measure on $\mathfrak{M}$, and

$$\int_X g\,d\varphi = \int_X gf\,d\mu$$

for every measurable g on X with range in $[0, \infty]$.

Note.
将 measurable function f 视为对 measure μ 的某种 countable 次算数加权操作. 那么得益于 μ 对 countable 操作的相容性, φ 也就成为一个测度. φ 的积分行为也验证了这种看法的合理性.

Proof.

- Let $E_1, \cdots, E_n, \cdots$ be disjoint measurable sets such that $E = \bigcup_{n=1}^{\infty} E_n$. Then $\left\{\sum_{k=1}^{n} \chi_{E_k} f\right\}$ be increasing measurable function sequence such that $\lim_{n\to\infty} \sum_{k=1}^{n} \chi_{E_k} f = f$. The Monotone Convergence Theorem shows that

$$\begin{aligned}\sum_{n=1}^{\infty} \varphi(E_k) = \lim_{n\to\infty} \sum_{k=1}^{n} \int_{E_k} f\,\mathrm{d}\mu &= \lim_{n\to\infty} \int_E \sum_{k=1}^{n} \chi_{E_k} f\,\mathrm{d}\mu \\ &= \int_E f\,\mathrm{d}\mu = \varphi(E)\end{aligned}$$

  Since $\varphi(\varnothing) = 0$. φ is a measure on X .

- To show the second results. See that it is true for $g = \chi_E$. Thus it is true for g is a simple function . And the general case follows from the Monotone Convergence Theorem.

□

2.5 Integration of Complex Functions

Definition 2.5.1 ▶ Lebegsgue Integrability

We define $L^1(\mu)$ to be the collection of all complex measurable functions f on X for which

$$\int_X |f|\,d\mu < \infty.$$

Note that the measurability of f implies that of $|f|$, hence the above integral is defined.
The members of $L^1(\mu)$ are called ***Lebesgue integrable functions*** (with respect to μ) or ***summable functions***.

Definition 2.5.2 ▸ Complex Integration

If $f = u + iv$, where u and v are real measurable functions on X, and if $f \in L^1(\mu)$, we define

$$\int_E f\, d\mu = \int_E u^+\, d\mu - \int_E u^-\, d\mu + i \int_E v^+\, d\mu - i \int_E v^-\, d\mu \tag{1}$$

for every measurable set E.

Remark.
We have $u^+ \leq |u| < |f|$, etc., so that each of these four integrals is finite. Thus (1) defines the integral on the left as a complex number.

Definition 2.5.3 ▸ Real Integration

Occasionally it is desirable to define the integral of a measurable function f with range in $[-\infty, \infty]$ to be

$$\int_E f\, d\mu = \int_E f^+\, d\mu - \int_E f^-\, d\mu, \tag{2}$$

provided that at least one of the integrals on the right of (2) is finite. The left side of (2) is then a number in $[-\infty, \infty]$.

Theorem 2.5.4

Suppose f and $g \in L^1(\mu)$ and α and β are complex numbers. Then $\alpha f + \beta g \in L^1(\mu)$, and

$$\int_X (\alpha f + \beta g)\, d\mu = \alpha \int_X f\, d\mu + \beta \int_X g\, d\mu$$

Proof sketch.
可积性来自于模长的三角不等式. 主要证明可加性. 将 $f+g=h$ 的正负部分到两边, 就都变成了非负函数. 积分即可.

Theorem 2.5.5

If $f \in L^1(\mu)$, then

$$\left|\int_X f d\mu\right| \le \int_X |f| d\mu.$$

Note.
被缩放的部分实际上是 f 因改变了方向产生的损耗.

Proof. Let $z = \int_X f \,\mathrm{d}\mu$, z is a complex number . There exists a complex number α ($\alpha = \frac{\bar{z}}{|z|}$) , such that $|\alpha| = 1$, and

$$\left|\int_X f \,\mathrm{d}\mu\right| = \alpha z = \alpha \int_X f \,\mathrm{d}\mu = \int_X \alpha f \,\mathrm{d}\mu$$

Suppose $\alpha f = u + iv$, since $\int_X \alpha f \,\mathrm{d}\mu$ is a real number , $\int_X v \,\mathrm{d}\mu = 0$. We have

$$\int_X \alpha f \,\mathrm{d}\mu = \int_X u \,\mathrm{d}\mu \le \int_X |\alpha f| \,\mathrm{d}\mu = \int_X |f| \,\mathrm{d}\mu$$

□

Theorem 2.5.6 ▸ Lebesgue's Dominated Convergence Theorem

Suppose $\{f_n\}$ is a sequence of complex measurable functions on X such that

$$f(x) = \lim_{n\to\infty} f_n(x)$$

exists for every $x \in X$. If there is a function $g \in L^1(\mu)$ such that

$$|f_n(x)| \le g(x) \quad (n = 1, 2, 3, \dots; x \in X),$$

then $f \in L^1(\mu)$,

$$\lim_{n\to\infty} \int_X |f_n - f| \,d\mu = 0,$$

and

$$\lim_{n\to\infty} \int_X f_n \, d\mu = \int_X f \, d\mu.$$

Proof sketch.
$\{f_n\}$ 不一定单调，能将极限从积分外塞进积分里的工具，目前我们只有 Fatou's Lemma，但是它原始的样子跟我们需要的不等号是反向的. 自然的想法是填一个负号，这又会导致函数失去非负性，而控制函数 g 恰好能弥补这一点不足. 实现起来，我们从 g 上扣去 f, f_n，而非直接拿出 f, f_n 去做.

Proof. Since $|f| \leq g$ and f is measurable, $f \in L^1(\mu)$. Function $(2g - |f_n - f|)$ is a nonnegative measurable function. Applying Fatou's Lemma to it, we have

$$\int_X \liminf_{n\to\infty} (2g - |f_n - f|) \, \mathrm{d}\mu \leq \liminf_{n\to\infty} \int_X (2g - |f_n - f|) \, \mathrm{d}\mu$$

where

$$\liminf_{n\to\infty} (2g - |f_n - f|) = 2g - \limsup_{n\to\infty} |f_n - f| = 2g$$

$$\liminf_{n\to\infty} \int_X (2g - |f_n - f|) \, \mathrm{d}\mu = 2\int_X g \, \mathrm{d}\mu - \limsup_{n\to\infty} \int_X |f_n - f| \, \mathrm{d}\mu$$

The original inequality comes to

$$2\int_X g \, \mathrm{d}\mu \leq 2\int_X g \, \mathrm{d}\mu - \limsup_{n\to\infty} \int_X |f_n - f| \, \mathrm{d}\mu$$

Since $\int_X g \, \mathrm{d}\mu$ is finite, we can substract $2\int_X g \, \mathrm{d}\mu$ both sides and get

$$\limsup_{n\to\infty} \int_X |f_n - f| \, \mathrm{d}\mu \leq 0$$

Thus

$$\lim_{n\to\infty} \int_X |f_n - f| \, \mathrm{d}\mu = 0$$

Finnaly , we have

$$\begin{aligned}\left|\lim_{n\to\infty}\int_X f_n \, \mathrm{d}\mu - \int_X f \, \mathrm{d}\mu\right| &= \left|\lim_{n\to\infty}\int_X (f_n - f) \, \mathrm{d}\mu\right| \\ &\leq \lim_{n\to\infty}\int_X |f_n - f| \, \mathrm{d}\mu \\ &= 0\end{aligned}$$

which completes the proof. □

2.6 The Sets of Measure Zero

Definition 2.6.1

Let P be a property which a point x may or may not have. If μ is a measure on a σ-algebra $\mathfrak{M}$ and if $E \in \mathfrak{M}$, the statement "P holds ***almost everywhere on*** E" (abbreviated to "P holds a.e. on E") means that there exists an $N \in \mathfrak{M}$ such that $\mu(N) = 0$, $N \subset E$, and P holds at every point of $E - N$.

Remark.
This concept of a.e. depends of course very strongly on the given measure, and we shall write "a.e. $[\mu]$" whenever clarity requires that the measure be indicated.

Example 2.6.2

For example, if f and g are measurable functions and if

$$\mu(\{x : f(x) \neq g(x)\}) = 0,$$

we say that $f = g$ a.e. $[\mu]$ on X, and we may write $f \sim g$.

Remark.
This is easily seen to be an equivalence relation. The transitivity ($f \sim g$ and $g \sim h$ implies $f \sim h$) is a consequence of the fact that the union of two sets of measure 0 has measure 0.

Proposition 2.6.3 ▶ Completion of a Measure

If $f \sim g$, then, for every $E \in \mathfrak{M}$,

$$\int_E f d\mu = \int_E g d\mu.$$

Theorem 2.6.4

Let $(X, \mathfrak{M}, \mu)$ be a measure space, let $\mathfrak{M}^*$ be the collection of all $E \subset X$ for which there exist sets A and $B \in \mathfrak{M}$ such that $A \subset E \subset B$ and $\mu(B - A) = 0$, and define $\mu(E) = \mu(A)$ in this situation. Then $\mathfrak{M}^*$ is a σ-algebra, and μ is a measure on $\mathfrak{M}^*$.

Remark.
This extended measure μ is called ***complete***, since all subsets of sets of measure 0 are now measurable; the σ-algebra $\mathfrak{M}^*$ is called the ***μ-completion*** of $\mathfrak{M}$. The theorem says that every measure can be completed, so, whenever it is convenient, we may assume that any given measure is complete; this just gives us more measurable sets, hence more measurable functions.

Proof. We begin by checking that μ is well defined for every $E \in \mathfrak{M}^*$. If for $i = 1, 2$, $A_i \subseteq E \subseteq B_i$, where $A_i, B_i \in \mathfrak{M}, \mu(B_i - A_i) = 0$. We need to show that $\mu(A_1) = \mu(A_2)$. Since $A_1 \subseteq E \subseteq B_2$,

$$\mu(A_1) \leq \mu(B_2) = \mu(A_2) + \mu(B_2 - A_2) = \mu(A_2)$$

By symmetric ,

$$\mu(A_2) \leq \mu(A_2)$$

Thus $\mu(A_1) = \mu(A_2)$, $\mu(E)$ is well defined. Next , let us verify that $\mathfrak{M}^*$ has the three defining properties of a σ-algebra.

1. It is clearly that $X \in \mathfrak{M}^*$.
2. If $A \subseteq E \subseteq B$, then $B^c \subseteq E^c \subseteq A^c$. And since $\mu(A^c - B^c) = \mu(A^c \cap B) = \mu(B - A) = 0$, $E^c \in \mathfrak{M}^*$.
3. Suppose $A_i \subseteq E_i \subseteq B_i, i = 1, 2, \cdots$, where $A_i, B_i \in \mathfrak{M}$, $\mu(B_i - A_i) = 0$.

Let $E = \bigcup E_i, A = \bigcup A_i, B = \bigcup B_i$. Since

$$B - A = \bigcup_i (B_i - A) \subseteq \bigcup_i (B_i - A_i)$$

,

$$\mu(B - A) \leq \sum_i \mu(B_i - A_i) = 0$$

Then by $A \subseteq E \subseteq B$, it follows that $E \in \mathfrak{M}^*$.

Finally, we show the countable additive of μ . If the above sets E_i are disjoint, the smae is true of the sets A_i , then

$$\mu(E) = \mu(A) = \sum_i \mu(A_i) = \sum_i \mu(E_i)$$

□

Definition 2.6.5 ▸ More Mearuable Functions

We say a function f defined on a set $E \in \mathfrak{M}$ is ***measurable on*** X . If $\mu(E^c) = 0$ and if $f^{-1}(V) \cap E$ is a measurable for every open set V .

Remark.

- If we define $f(x) = 0$ for $x \in E^c$, we obtain a mearuable function on X , in the old sense.
- If our measure happens to be complete , we can define f on E^c in a perfrctly arbitrary manner, and we still get a measurable function . The integral of f over any set $A \in \mathfrak{M}$ is independent of the definition of f on E^c; therefore this definition need not even be specified at all.

Theorem 2.6.6

Suppose $\{f_n\}$ is a sequence of complex measurable functions defined a.e. on X such that

$$\sum_{n=1}^{\infty} \int_X |f_n| \, d\mu < \infty. \tag{1}$$

Then the series

$$f(x) = \sum_{n=1}^{\infty} f_n(x) \tag{2}$$

converges for almost all x, $f \in L^1(\mu)$, and

$$\int_X f \, d\mu = \sum_{n=1}^{\infty} \int_X f_n \, d\mu. \tag{3}$$

Remark.

- 现在, 考察积分和求和是否可交换, 只需要检查模长积分的级数是否收敛.
- 即便 f_n 在 X 上处处被定义, 级数收敛的地方也只是 ***almost everywhere*** .

Note.
此时, 控制函数就是绝对级数本身, 它在一个满测集上将原级数控制住. 其特殊性间接给出了原级数的几乎处处收敛性质.

Proof. Let S_n be the set on which f_n is defined, so that $\mu\left(S_n^c\right) = 0$. Let $S = \bigcap S_n$. Put $\varphi\left(x\right) = \sum \left|f_n\left(x\right)\right|$, for $x \in S$ .By Monotone Convergence Theorem

$$\int_S \varphi \, \mathrm{d}\mu \le \sum_{n=1}^{\infty} \int_S \left|f_n\right| \, \mathrm{d}\mu < \infty$$

Let $E = \{x \in S : \varphi\left(x\right) < \infty\}$. Then it follows that $\mu\left(E^c\right) = 0$. We have

$$\sum_{n=1}^{\infty} \left|f_n\left(x\right)\right| < \infty, \quad \forall x \in E$$

Thus

$$f\left(x\right) = \sum_{n=1}^{\infty} f_n\left(x\right)$$

converges on E , which is a set with its complement measure zero . That is to say f coverges for almost all x . And since $f \in L^1\left(\mu|_E\right)$, $\mu\left(E\right) = 0$, we have

$f \in L^1(\mu)$. Finally , by Monotone Convergence Theorem ,

$$\int_X f \, \mathrm{d}\mu = \int_E f \, \mathrm{d}\mu = \sum_{n=1}^{\infty} \int_E f_n \, \mathrm{d}\mu = \sum_{n=1}^{\infty} \int_X f_n \, \mathrm{d}\mu$$

□

Theorem 2.6.7

1. Suppose $f : X \to [0, \infty]$ is measurable , $E \in \mathfrak{M}$, and $\int_E f \, \mathrm{d}\mu = 0$. Then $f = 0$ a.e. on E.
2. Suppose $f \in L^1(\mu)$ and $\int_E f \, \mathrm{d}\mu = 0$ for every $E \in \mathfrak{M}$. Then $f = 0$ a.e. on X .
3. Suppose $f \in L^1(\mu)$ and

$$\left| \int_X f \, \mathrm{d}\mu \right| = \int_X |f| \, \mathrm{d}\mu$$

Then there is a constant α such that $\alpha f = |f|$ *a.e.* on X .

Note.
3. 这个积分等式的意义是不存在积分项因方向改变产生的损耗. 结论就是说如果这种损耗不存在, 那么 f 几乎不改变方向.

Proof. 1. Let $A_n = \left\{ x \in E : f(x) \geq \frac{1}{n} \right\}$. Then

$$0 = \int_E f \, \mathrm{d}\mu \geq \int_{A_n} f \, \mathrm{d}\mu \geq \mu(A_n) \frac{1}{n}.$$

It follows that

$$\mu(A_n) = 0, \quad \forall n = 1, 2, \cdots$$

Let $A = \bigcup_{n=1}^{\infty} A_n$. Then $\mu(A) = 0$, and $f \equiv 0$ on $E \setminus A$. Thus $f = 0$ a.e. on E .

2. Suppose $f = u + iv$. Let $E = \{x \in X : u(x) \geq 0\}$. Since $\operatorname{Re} f = u^+$ on E , then

$$\int_E u^+ \, \mathrm{d}\mu = 0$$

It follows by 1. that $\mu^+ = 0$ a.e. on E . Since $\mu^+(X \setminus E) \equiv 0$, $\mu^+ = 0$ a.e.

on X . We conclude similarly that

$$u^- = v^+ = v^- = 0, \quad a.e.$$

3. Let $z = \int_X f \, \mathrm{d}\mu$. Then z is a complex number . Take $\alpha = \bar{z}/ |z|$ .Then

$$\alpha z = |z| = \left| \int_X f \, \mathrm{d}\mu \right|$$

Where

$$\alpha z = \alpha \int_X f \, \mathrm{d}\mu = \int_X \alpha f \, \mathrm{d}\mu$$

It follows that

$$\int_X \alpha f \, \mathrm{d}\mu = \left| \int_X f \, \mathrm{d}\mu \right| = \int_X |f| \, \mathrm{d}\mu \implies \int_X (\alpha f - |f|) \, \mathrm{d}\mu = 0$$

which shows that

$$\alpha f = |f| , \quad a.e. \text{ on } X$$

□

Theorem 2.6.8

Suppose $\mu(X) < \infty$, $f \in L^1(\mu)$, S is a closed set in the complex plane, and the averages

$$A_E(f) = \frac{1}{\mu(E)} \int_E f \, d\mu$$

lie in S for every $E \in \mathfrak{M}$ with $\mu(E) > 0$. Then $f(x) \in S$ for almost all $x \in X$.

Proof. Let Δ be circular disk , with centre α and radius r . And suppose that Δ lies in S^c . S^c can be written as the union of countably many such disks . Thus, we only need to show that $\mu(E) = 0$, where $E = f^{-1}(\Delta)$. If $\mu(E) > 0$, then

$$|A_E(f) - \alpha| = \frac{1}{\mu(E)} \left| \int_E (f - \alpha) \, \mathrm{d}\mu \right| \le \frac{1}{\mu(E)} \int_E |f - \alpha| \, \mathrm{d}\mu \le r$$

Which leads to a contradition, since $A_E(f) \in S$. Hence $\mu(E) = 0$. □

Theorem 2.6.9

Let $\{E_k\}$ be a sequence of measurable sets in X, such that

$$\sum_{k=1}^{\infty} \mu(E_k) < \infty.$$

Then almost all $x \in X$ lie in at most finitely many of the sets E_k.

Proof. Let

$$g(x) = \sum_{k=1}^{\infty} \chi_{E_k}(x)$$

Then g is measurable . By Monotone Convergence Theorem ,

$$\int_X g(x)\, \mathrm{d}\mu = \sum_{k=1}^{\infty} \int_X \chi_{E_k}(x)\, \mathrm{d}\mu = \sum_{k=1}^{\infty} \mu(E_k) < \infty$$

Thus $g \in L^1(\mu)$. There is a $E \in \mathfrak{M}$, $\mu(E^c) = 0$, such that $g^{-1}(\infty) \cap E = \varnothing$. If $x \in X$ lie in infinitely many of the sets E_k . Then $g(x) = \infty$, it follows that $x \in E^c$. Thus , points that lie in infinitely many of the sets E_k will always lie in E^c as well , whose measure is zero . □

2.7 Exersices

Note.
关于积分的问题有两种常用标准技巧. 一种是按取值划分集合. 一种用可控的函数列逼近.

Problem 2.7.1

Does there exist an infinite σ-algebra which has only countably many members?

Proof. □

Problem 2.7.2

Prove that if f is a real function on a measurable space X such that $\{x : f(x) \geq r\}$ is measurable for every rational r, then f is measurable.

Proof. Take any real number a . We have

$$f^{-1}((a, \infty]) = \bigcup_{r \in \mathbb{Q}, r > a} f^{-1}([r, \infty])$$

which is a countably many union of measurable sets . Thus $f^{-1}((a, \infty])$ is measurable for all a . It follows that f is measurable . □

Problem 2.7.3

Let $\{a_n\}$ and $\{b_n\}$ be sequences in $[-\infty, \infty]$, and prove the following assertions:

(a)

$$\limsup_{n\to\infty}(-a_n) = -\liminf_{n\to\infty} a_n.$$

(b)

$$\limsup_{n\to\infty}(a_n + b_n) \leq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n$$

provided none of the sums is of the form $\infty - \infty$.

(c) If $a_n \leq b_n$ for all n, then

$$\liminf_{n\to\infty} a_n \leq \liminf_{n\to\infty} b_n.$$

Show by an example that strict inequality can hold in (b).

Proof. 1.

$$\limsup_{n\to\infty}(-a_n) = \lim_{n\to\infty} \sup_{k\geq n}\{-a_k\} = -\lim_{n\to\infty} \inf_{k\geq n}\{a_k\} = -\liminf_{n\to\infty} a_n$$

2.
$$\begin{aligned}\limsup_{n\to\infty}(a_n+b_n) &= \lim_{n\to\infty}\sup_{k\geq n}\{a_k+b_k\}\\ &\leq \lim_{n\to\infty}\left(\sup_{k\geq n}\{a_k\}+\sup_{k\geq n}\{b_k\}\right)\\ &= \lim_{n\to\infty}\sup_{k\geq n}\{a_k\}+\lim_{n\to\infty}\sup_{k\geq n}\{b_k\}\\ &= \limsup_{n\to o} a_n+\limsup_{n\to\infty} b_n\end{aligned}$$

Let $\{a_n\}=\left\{(-1)^n\right\}, \{b_n\}=\left\{(-1)^{n+1}\right\}$. Then the RHS of the inequality is $1+1=2$. The LHS is 0 .

3. If $a_n \leq b_n$ for all n , then
$$\inf_{k\geq n}\{a_k\}\leq \inf_{k\geq n}\{b_k\}$$
Thus
$$\liminf_{n\to\infty} a_n=\lim_{n\to\infty}\inf_{k\geq n}\{a_k\}\leq \lim_{n\to\infty}\inf_{k\geq n}\{b_k\}=\liminf_{n\to\infty} b_n$$

□

Problem 2.7.4

(a) Suppose $f: X\to[-\infty,\infty]$ and $g: X\to[-\infty,\infty]$ are measurable. Prove that the sets
$$\{x: f(x)<g(x)\},\{x: f(x)=g(x)\}$$
are measurable.
(b) Prove that the set of points at which a sequence of measurable real-valued functions converges (to a finite limit) is measurable.

Proof. 1. Let
$$h=\begin{cases} g-f, & x\in f^{-1}((-\infty,\infty))\\ 0, & f(x)=g(x)=-\infty\\ 0, & f(x)=g(x)=\infty\\ -1, & f(x)=\infty, g(x)<\infty\\ 1, & f(x)=-\infty, g(x)>-\infty\end{cases}$$

It is easy to see that h is measurable. Thus

$$\{x : f(x) < g(x)\} = h^{-1}((0, \infty])$$

is measruable . And

$$\{x : f(x) = g(x)\} = h^{-1}(0)$$

is measurable .

2. Let $\{f_n\}$ be a sequence of measurable real-valued functions . Take $x \in X$. $\{f_n(x)\}$ is converges. If and only if for any $k \in \mathbb{N}$, there exists $N \in \mathbb{N}$, such that for all $m, n \geq N$, $|f_m(x) - f_n(x)| \leq \frac{1}{k}$. That is to say

$$\{x : \{f_n(x)\} \text{ converges }\} = \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{m=N}^{\infty} \bigcap_{n=N}^{\infty} (|f_m - f_n|)^{-1} \left((-\infty, \frac{1}{k}]\right)$$

which is a measurable set .

□

Problem 2.7.5

Let X be an uncountable set, let $\mathfrak{M}$ be the collection of all sets $E \subset X$ such that either E or E^c is at most countable, and define $\mu(E) = 0$ in the first case, $\mu(E) = 1$ in the second. Prove that $\mathfrak{M}$ is a σ-algebra in X and that μ is a measure on $\mathfrak{M}$. Describe the corresponding measurable functions and their integrals.

Proof. We will first show $\mathfrak{M}$ is a σ-algebra step by step .

1. It is clear that if $E \in \mathfrak{M}$, E^c is as well .
2. Let $E_1, E_2, \cdots$ be members of $\mathfrak{M}$. If all of E_n is countable , then $\bigcup_{n=1}^{\infty} E_n$ is at most countable. Otherwise ,we may assume that E_1 is uncountable ,then E^c is at most countable.

$$X \setminus \bigcup_{n=1}^{\infty} E_n = E_1^c \cap \bigcap_{n=2}^{\infty} E_n^c$$

is at most countable . Hence $\bigcup_{n=1}^{\infty} E_n \in \mathfrak{M}$.

3. Since $\varnothing$ is at most coutable , $X \in \mathfrak{M}$.

The above shows that $\mathfrak{M}$ is a σ-algebra . To see μ is a measure , take $E_1, E_2, \cdots \in \mathfrak{M}$ with any two of them have empty intersection . If all of E_n is countable , then $\bigcup_{n=1}^{\infty} E_n$ is as well . We have

$$\sum_{n=1}^{\infty} \mu(E_n) = 0 = \mu\left(\bigcup_{n=1}^{\infty} E_n\right)$$

If one of the sets E_n is uncountable , say E_1 . If $E_i, i \geq 2$ is uncountable.

$$X = (\varnothing)^c = (E_1 \cap E_2)^c = E_1^c \cup E_2^c$$

is at most countable , which leads to a contradiction . Thus $E_i, i \geq 2$ is at most countable. Hence

$$\sum_{n=1}^{\infty} \mu(E_n) = 1 + 0 + \cdots + 0 = 1 = \mu\left(\bigcup_{n=1}^{\infty} E_n\right).$$

It follows that μ is a measure on $\mathfrak{M}$, since $\mu(\varnothing) = 0$.

To describe the measurable functions . If f is not *a.e.* constant , then there are two value a, b of f , such that $f^{-1}(a)$, $f^{-1}(b)$ is uncountable . Then

$$f^{-1}(b) \subseteq \left(f^{-1}(a)\right)^c$$

is at most countable , which leads to a contradiction. Thus f is *a.e.* constant. Take any measurable function f . Suppose that $E = f^{-1}(c)$, $\mu(E) \neq 0$. Then $\mu(E^c) = 0$. It follows that f is integrable and

$$\int_X f \,\mathrm{d}\mu = \int_E f \,\mathrm{d}\mu = c\mu(E) = c$$

□

Problem 2.7.6

Suppose $f_n : X \to [0, \infty]$ is measurable for $n = 1, 2, 3, \ldots$, $f_1 \geq f_2 \geq f_3 \geq \cdots \geq 0$, $f_n(x) \to f(x)$ as $n \to \infty$, for every $x \in X$, and $f_1 \in L^1(\mu)$. Prove that then

$$\lim_{n\to\infty} \int_X f_n \, d\mu = \int_X f \, d\mu$$

and show that this conclusion does not follow if the condition "$f_1 \in L^1(\mu)$" is omitted.

Proof. Let $g_n = f_1 - f_n$, then

$$0 = g_1 \leq g_2 \leq \cdots, \quad g_n(x) \to f_1(x) - f(x) \text{ as } n \to \infty$$

Applying the Monotone Convergence Theorem to $\{g_n\}$, we have

$$\int_X f_1 \, d\mu - \lim_{n\to\infty} \int_X f_n \, d\mu = \lim_{n\to\infty} \int_X g_n \, d\mu = \int_X (f_1 - f) \, d\mu = \int_X f_1 \, d\mu - \int_X f \, d\mu$$

Since $\int_X f_1 \, d\mu < \infty$, substracting $\int_X f_1 \, d\mu$ both sides , we have

$$\lim_{n\to\infty} \int_X f_n \, d\mu = \int_X f \, d\mu.$$

Let $X = \mathbb{R}$ with borel measure, and let $f_n = \chi_{\mathbb{R}\setminus[0,n]} \frac{1}{n}$. Then $f_n(x) \to f(x) = 0$, $\int_X f \, d\mu = 0$. But

$$\int_X f_n \, d\mu = \infty, \quad \forall n$$

Thus the condition "$f_1 \in L^1(\mu)$" can not be omitted. □

Problem 2.7.7

Put $f_n = \chi_E$ if n is odd, $f_n = 1 - \chi_E$ if n is even. What is the relevance of this example to Fatou's lemma?

Proof.

$$\liminf_{n\to\infty} f_n(x) \equiv 0, \qquad \int_X \liminf_{n\to\infty} f_n \, d\mu = 0$$

$$\int_X f_n \, d\mu = \begin{cases} \mu(E), & n \text{ is odd} \\ \mu(X \setminus E), & n \text{ is even} \end{cases}$$

$$\liminf_{n\to\infty} \int_X f_n \, d\mu = \min\{\mu(E), \mu(X \setminus E)\}$$

□

Problem 2.7.8

Suppose μ is a positive measure on X, $f : X \to [0, \infty]$ is measurable, $\int_X f\, d\mu = c$, where $0 < c < \infty$, and α is a constant. Prove that

$$\lim_{n\to\infty} \int_X n \log[1 + (f/n)^\alpha]\, d\mu = \begin{cases} \infty & \text{if } 0 < \alpha < 1, \\ c & \text{if } \alpha = 1, \\ 0 & \text{if } 1 < \alpha < \infty. \end{cases}$$

Hint: If $\alpha \geq 1$, the integrands are dominated by αf. If $\alpha < 1$, Fatou's lemma can be applied.

Proof. By differenting $\frac{1}{y} \log(1 + ky)$. We know $n \log\left(1 + \frac{f}{n}\right)$ is increasing as n is increasing. By Monotone Convergence Theorem ,

$$\lim_{n\to\infty} \int_X n \log\left[1 + \left(\frac{f}{n}\right)\right] \mathrm{d}\mu = \int_X \lim_{n\to\infty} n \log\left[1 + \frac{f}{n}\right] \mathrm{d}\mu = \int_X f\, \mathrm{d}\mu = c$$

. If $\alpha > 1$, then note that

$$0 \leq n \log\left[1 + \left(\frac{f}{n}\right)^\alpha\right] \leq \alpha f, \quad \int_X \alpha f\, \mathrm{d}\mu = \alpha c < \infty$$

The Dominated Convergence Theorem gives that

$$\lim_{n\to\infty} \int_X n \log\left[1 + \left(\frac{f}{n}\right)^\alpha\right] \mathrm{d}\mu = \int_X \lim_{n\to\infty} n \log\left[1 + \left(\frac{f}{n}\right)^\alpha\right] \mathrm{d}\mu = 0$$

If $\alpha < 1$, the Fatou;s Lemma gives that

$$\lim_{n\to\infty} \int_X n \log\left[1 + \left(\frac{f}{n}\right)^\alpha\right] \mathrm{d}\mu \geq \int_X \lim_{n\to\infty} n \log\left[1 + \left(\frac{f}{n}\right)^\alpha\right] \mathrm{d}\mu = \int_X \infty\, \mathrm{d}\mu = \infty$$

□

Problem 2.7.9

Suppose $\mu(X) < \infty$, $\{f_n\}$ is a sequence of bounded complex measurable functions on X, and $f_n \to f$ uniformly on X. Prove that

$$\lim_{n\to\infty} \int_X f_n \, d\mu = \int_X f \, d\mu,$$

and show that the hypothesis "$\mu(X) < \infty$" cannot be omitted.

Proof. Take any $\varepsilon > 0$,

$$\lim_{n\to\infty} \int_X |f_n - f| \, \mathrm{d}\mu \le \mu(X)\,\varepsilon$$

Let $\varepsilon \to 0$,we have

$$\lim_{n\to\infty} \int_X |f_n - f| \, \mathrm{d}\mu = 0$$

Thus

$$\left| \lim_{n\to\infty} \int_X f_n \, \mathrm{d}\mu - \int_X f \, \mathrm{d}\mu \right| \le \lim_{n\to\infty} \int_X |f_n - f| \, \mathrm{d}\mu = 0$$

Let $f_n = \chi_{[0,n]} \frac{1}{n}$. Then $f = 0$,

$$\lim_{n\to\infty} \int_X f_n \, \mathrm{d}\mu = 1$$

$$\int_X f \, \mathrm{d}\mu = 0$$

Thus $\mu(X) < \infty$ cannot be omitted. □

Problem 2.7.10

Let $\{E_k\}$ be a sequence of measurable sets in X, such that

$$\sum_{k=1}^{\infty} \mu(E_k) < \infty.$$

Suppose A is the set of all x which lie in infinitely many E_k . Show that

$$A = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k$$

Then almost all $x \in X$ lie in at most finitely many of the sets E_k.

Proof. $x \in A^c$ iff there exists $k \in \mathbb{N}$, such that for all $n \geq k$, $x \in E_k^c$. That is

$$A^c = \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} E_k^c$$

Thus

$$A = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} E_k$$

$$\mu(A) = \lim_{k\to\infty} \mu\left(\bigcup_{n=k}^{\infty} E_k\right) \leq \lim_{k\to\infty} \sum_{n=k}^{\infty} \mu(E_k) = \left(\sum_{k=1}^{\infty} \mu(E_k) - \lim_{k\to\infty} \sum_{m=1}^{k} \mu(E_k)\right) = 0$$

□

Problem 2.7.11

Suppose $f \in L^1(\mu)$. Prove that to each $\epsilon > 0$ there exists a $\delta > 0$ such that $\int_E |f|\, d\mu < \epsilon$ whenever $\mu(E) < \delta$.

Proof. Let

$$f_n := \min\{|f|, n\}$$

Then

$$0 \leq f_1 \leq f_2 \leq \cdots, \quad f_n \to |f| \text{ as } n \to \infty$$

$$\int_E |f|\, \mathrm{d}\mu \leq \lim_{n\to\infty} \int_X f_n\, \mathrm{d}\mu$$

There exists $N \in \mathbb{N}$,such taht

$$\int_E |f|\, \mathrm{d}\mu \leq \int_E f_N\, \mathrm{d}\mu + \frac{\varepsilon}{2} \leq \mu(E) N + \frac{\varepsilon}{2} \leq \varepsilon$$

whenever $\mu(E) < \delta := \frac{\varepsilon}{2N}$ □

Problem 2.7.12

If $f \geq 0$, then

$$\int_E \infty \cdot f \, d\mu = \infty \cdot \int_E f \, d\mu.$$

Proof. Let $E_1 = f^{-1}(0) \cap E$, $E_2 = E \setminus E_1$.

1. If $\mu(E_2) > 0$ Then

$$\int_E \infty \cdot f \, \mathrm{d}\mu \geq \int_{E_2} \infty \cdot f \, \mathrm{d}\mu = \infty$$

$$\int_E f \, \mathrm{d}\mu \geq \int_{E_2} f \, \mathrm{d}\mu > 0, \quad \infty \cdot \int_E f \, \mathrm{d}\mu = \infty$$

2. If $\mu(E_2) = 0$, then

$$\int_E \infty \cdot f \, \mathrm{d}\mu = \int_{E_1} \infty \cdot 0 \, \mathrm{d}\mu + \int_{E_2} \infty \cdot f \, \mathrm{d}\mu = 0$$

$$\int_E f \, \mathrm{d}\mu = \int_{E_1} 0 \, \mathrm{d}\mu + \int_{E_2} f \, \mathrm{d}\mu = 0, \quad \infty \cdot \int_E f \, \mathrm{d}\mu = 0$$

□

3 Measure Theory

3.1 Outer Measure

Definition 3.1.1 ▸ Outer Measure

A function φ defined for every subset A of an arbitrary set X is called an ***outer measure*** on X if the following conditions are satisfied:

(i) $\varphi(\varnothing) = 0$,
(ii) $0 \leq \varphi(A) \leq \infty$ whenever $A \subset X$,
(iii) $\varphi(A_1) \leq \varphi(A_2)$ whenever $A_1 \subset A_2$,
(iv) $\varphi(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \varphi(A_i)$ for every countable collection of sets $\{A_i\}$ in X.

Example 3.1.2 ▸ Outer Measure

(i) In an arbitrary set X, define $\varphi(A) = 1$ if A is nonempty and $\varphi(\varnothing) = 0$.
(ii) Let $\varphi(A)$ be the number (possibly infinite) of points in A.
(iii) Let $\varphi(A) = \begin{cases} 0 & \text{if card } A \leq \aleph_0, \\ 1 & \text{if card } A > \aleph_0. \end{cases}$
(iv) If X is a metric space, fix $\varepsilon > 0$. Let $\varphi(A)$ be the smallest number of balls of radius ε that cover A.
(v) Select a fixed x_0 in an arbitrary set X, and let

$$\varphi(A) = \begin{cases} 0 & \text{if } x_0 \notin A, \\ 1 & \text{if } x_0 \in A; \end{cases}$$

φ is called the ***Dirac measure*** concentrated at x_0.

Definition 3.1.3

Let φ be an outer measure on a set X. A set $E \subset X$ is called φ-***measurable*** if

$$\varphi(A) = \varphi(A \cap E) + \varphi(A - E)$$

for every set $A \subset X$. In view of property (iv) above, observe that φ-measurability requires only

$$\varphi(A) \geq \varphi(A \cap E) + \varphi(A - E).$$

Lemma 3.1.4

A set $E \subset X$ is φ-measurable if and only if

$$\varphi(P \cup Q) = \varphi(P) + \varphi(Q)$$

for all sets P and Q such that $P \subset E$ and $Q \subset E^c$.

Proof. For Necessity, if E is φ-measurable , then for all such P, Q,

$$\varphi\left(P \cup Q\right) = \varphi\left(\left(P \cup Q\right) \cap E\right) + \varphi\left(\left(P \cup Q\right) - E\right) = \varphi\left(P\right) + \varphi\left(Q\right)$$

For sufficiency , let $P = A \cap E$, $Q = A \cap E^c = A \setminus E$. Then

$$\varphi\left(A\right) = \varphi\left(P \cup Q\right) = \varphi\left(P\right) + \varphi\left(Q\right) = \varphi\left(A \cap E\right) + \varphi\left(A \setminus E\right)$$

□

Theorem 3.1.5

Suppose φ is an outer measure on an arbitrary set X. Then the following four statements hold:

(i) E is φ-measurable whenever $\varphi(E) = 0$.
(ii) $\varnothing$ and X are φ-measurable.
(iii) $E_1 - E_2$ is φ-measurable whenever E_1 and E_2 are φ-measurable.
(iv) If $\{E_i\}$ is a ***countable collection of disjoint φ-measurable sets***, then $\bigcup_{i=1}^{\infty} E_i$ is φ-measurable and

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \varphi(E_i).$$

More generally, if $A \subset X$ is an arbitrary set, then

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c),$$

where

$$S = \bigcup_{i=1}^{\infty} E_i.$$

Proof. 1. Take any $A \subseteq X$. Since $A \cap E \subseteq E$, $\varphi(A \cap E) = 0$. Thus

$$\varphi(A \cap E) + \varphi(A \cap E^c) = \varphi(A \cap E^c) \leq \varphi(A)$$

2. Follows immediately frome lemma 3.1.4.

3. Take any P, Q such that $P \subseteq (E_1 \setminus E_2) = E_1 \cap E_2^c$, $Q \subseteq (E_1 \setminus E_2)^c = E_1^c \cup E_2$.

$$\varphi(Q) = \varphi(Q \cap E_2) + \varphi(Q - E_2)$$

Note that $Q - E_2 \subseteq E_1^c$, $Q \cap E_2 \subseteq E_2$. Since $P \subseteq E_1$, the measurability of E_1 implis that

$$\varphi(P) + \varphi(Q - E_2) = \varphi(P \cup (Q - E_2))$$

Note that $P \cup (Q - E_2) \subseteq E_2^c$. Since $Q \cap E_2 \subseteq E_2$, we have

$$\varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) = \varphi(P \cup Q)$$

Finally , we have

$$\begin{aligned} \varphi(P) + \varphi(Q) &= \varphi(P) + \varphi(Q \cap E_2) + \varphi(Q - E_2) \\ &= \varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) \\ &= \varphi(P \cup Q) \end{aligned}$$

4. Let $S_k = \bigcup_{i=1}^{k} E_i$. We proceed by finite induction and first note that the result is obviously true for $k = 1$. For $k > 1$ assume that S_k is φ-measurable and that

$$\varphi(A) \geq \sum_{i=1}^{k} \varphi(A \cap E_i) + \varphi\left(A \cap S_k^c\right), \quad \forall A \subseteq X$$

Then

$$\begin{aligned}\varphi(A) &= \varphi(A \cap E_{k+1}) + \varphi\left(A \cap E_{k+1}^c\right) \\ &= \varphi(A \cap E_{k+1}) + \varphi\left(A \cap E_{k+1}^c \cap S_k^c\right) + \varphi\left(A \cap E_{k+1}^c \cap S_k\right) \\ &= \varphi(A \cap E_{k+1}) + \varphi\left(A \cap S_{k+1}^c\right) + \varphi(A \cap S_k)\end{aligned}$$

Replaced A by $A \cap S_k$, we have

$$\varphi(A \cap S_k) \geq \sum_{i=1}^{k} \varphi(A \cap S_k \cap E_i) + \varphi\left(A \cap S_k \cap S_k^c\right) = \sum_{i=1}^{k} \varphi(A \cap E_i)$$

Thus

$$\varphi(A) \geq \sum_{i=1}^{k+1} \varphi(A \cap E_{k+1}) + \varphi\left(A \cap S_{k+1}^c\right)$$

Then the subadditivity gives that

$$\varphi(A) \geq \varphi(A \cap S_{k+1}) + \varphi\left(A \cap S_{k+1}^c\right)$$

this , inturn , implies that S_{k+1} is φ-measurable. Since we now know for every set $A \subseteq X$ and for all positive integers k

$$\varphi(A) \geq \sum_{i=1}^{k} \varphi(A \cap E_i) + \varphi\left(A \cap S_k^c\right)$$

We have

$$\varphi(A) \geq \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c) \geq \varphi(A \cap S) + \varphi(A \cap S^c)$$

where $S = \bigcup_{i=1}^{\infty} E_k$. Thus S is measurable. The subadditivity gives that

$$\varphi(A) \leq \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c)$$

Finally, we have

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c)$$

Replace A by S , we can get

$$\varphi(S) = \sum_{i=1}^{\infty} \varphi(E_i)$$

□

Lemma 3.1.6

Let $\{E_i\}$ be a sequence of arbitrary sets. Then there exists a sequence of disjoint sets $\{A_i\}$ such that each $A_i \subset E_i$ and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i.$$

If each E_i is φ-measurable, and so is A_i.

Proof. Let $S_k = \bigcup_{i=1}^{k} E_i$. Let $A_k = S_k \setminus S_{k-1}, k \geq 1$, where $S_0 = \varnothing$. Then

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} (S_k \setminus S_{k-1}) = \bigcup_{i=1}^{\infty} A_i.$$

If each E_i is φ-measurable. $E_{i+1} \cup E_i = (E_{i+1} - E_i) \cup E_i$ is measurable follows from the (ii), (iii) of the above theorem . Inductively , every $S_k = (S_{k-1} \setminus E_k) \cup E_k$ is measurable. Then each A_k is measurable . □

Theorem 3.1.7

If $\{E_i\}$ is a sequence of φ-***measurable*** sets in X, then $\cup_{i=1}^{\infty} E_i$ and $\cap_{i=1}^{\infty} E_i$ are φ-***measurable***.

Proof. By the lemma above, there exists a sequence of disjoint φ - measurable sets $\{A_i\}$ such that each $A_i \subseteq E_i$, and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i$$

It follows from Theorem **??** that $\bigcup_{i=1}^{\infty} A_i$ is measurable , then $\bigcup_{i=1}^{\infty} E_i$ is as well . Since X is measurable , $E_i^c = X \setminus E_i$ is measurable , then $\bigcup_{i=1}^{\infty} E_i^c$ is measurable . Thus

$$\bigcap_{i=1}^{\infty} E_i = X \setminus \bigcup_{i=1}^{\infty} E_i^c$$

is measurable . □

Definition 3.1.8

A nonempty collection Σ of sets E satisfying the following two conditions is called a ***σ-algebra***:

(i) if $E \in \Sigma$, then $E^c \in \Sigma$;

(ii) $\cup_{i=1}^{\infty} E_i \in \Sigma$ if each E_i is in Σ.

Definition 3.1.9

In a topological space, the elements of the smallest σ-algebra that contains all open sets are called ***Borel sets***. The term "smallest" is taken in the sense of inclusion.

Proof. We need to show that the "smallest" exists. □

Corollary 3.1.10

If φ is an ***outer measure*** on an arbitrary set X, then the class of φ-measurable sets forms a ***σ-algebra***.

Proof. Follows immediately from Theorem **??** and Theorem **??**. □

Corollary 3.1.11

Suppose φ is an outer measure on X and $\{E_i\}$ a countable collection of φ-measurable sets.

(i) If $E_1 \subset E_2$ with $\varphi(E_1) < \infty$, then

$$\varphi(E_2 \setminus E_1) = \varphi(E_2) - \varphi(E_1).$$

(ii) (Countable additivity) If $\{E_i\}$ is a disjoint sequence of sets, then

$$\varphi(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \varphi(E_i).$$

(iii) (Continuity from the left) If $\{E_i\}$ is an increasing sequence of sets, that is, if $E_i \subset E_{i+1}$ for each i, then

$$\varphi(\bigcup_{i=1}^{\infty} E_i) = \varphi(\lim_{i\to\infty} E_i) = \lim_{i\to\infty} \varphi(E_i).$$

(iv) (Continuity from the right) If $\{E_i\}$ is a decreasing sequence of sets, that is, if $E_i \supset E_{i+1}$ for each i, and if $\varphi(E_{i_0}) < \infty$ for some i_0, then

$$\varphi(\bigcap_{i=1}^{\infty} E_i) = \varphi(\lim_{i\to\infty} E_i) = \lim_{i\to\infty} \varphi(E_i).$$

(v) If $\{E_i\}$ is any sequence of φ-measurable sets, then

$$\varphi(\liminf_{i\to\infty} E_i) \leq \liminf_{i\to\infty} \varphi(E_i).$$

(vi) If

$$\varphi(\bigcup_{i=i_0}^{\infty} E_i) < \infty$$

for some positive integer i_0, then

$$\varphi(\limsup_{i\to\infty} E_i) \geq \limsup_{i\to\infty} \varphi(E_i).$$

Summer Exercises

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